Spectral–Spatial Adversarial Multidomain Synthesis Network for Cross-Scene Hyperspectral Image Classification

Xi Chen, Lin Gao, Maojun Zhang, Chen Chen, and Shen Yan

***Abstract*— Cross-scene hyperspectral image (HSI) classification has received widespread attention due to its practicality. However, domain adaptation (DA)-based cross-scene HSI classification methods are typically tailored for a specific target scene involved in model training and require retraining for new scenes. We instead propose a novel spectral–spatial adversarial multidomain synthetic network (S^2AMSnet) that can be trained on a single source domain (SD) and generalized to unseen domains. S^2AMSnet improves the robustness of the model to the unseen domain by expanding the diverse distribution of the SD. Specifically, to spatially and spectrally generate diversified generative domain (GD), the spectral–spatial domain-generation network (S^2DGN) is designed, and two S^2DGNs with the same structure but not shared parameters are enabled to generate diversified GD through two-step min–max strategy. A Multidomain mixing module is employed to expand the diversity of the GD further and enhance their class-domain semantic consistency information. In addition, a multiscale mutual information (MI) regularization network (MMIRN) is used to constrain the S^2DGN so that the intrinsic class semantic information of its generated GD does not deviate from the SD. A semantic consistency discriminator with spectral–spatial feature extraction capability is utilized to capture class-domain semantic consistency information from diverse GD to obtain cross-domain invariant knowledge. Comparative analysis with eight state-of-the-art transfer learning methods on three real HSI datasets, along with an ablation study, validates the effectiveness of the proposed S^2AMSnet in the cross-scene HSI classification task. The codes of this work will be available at https://github.com/daxichen/S2AMSnet.**

***Index Terms*— Cross-scene classification, hyperspectral images (HSIs), mutual information (MI) regularization, single-source domain (SD) generalization.**

I. Introduction

HYPERSPECTRAL remote sensing is a multidimensional information acquisition technology that combines imaging and spectroscopic technologies, which can both imaging and spectroscopy [1]. Hyperspectral images (HSIs) obtained by imaging spectrometer are characterized by the unity of image and spectrum [2], and each of its pixels corresponds to a near-continuous spectral curve [3], which can respond to the diagnostic spectral absorption differences of the ground objects. Benefiting from the abundant spatial and spectral information provided by HSIs [4], the fine classification of materials using HSIs has been widely used in the fields of wetland monitoring [5], [6], precision agriculture [7], [8], urban planning [9], [10], and resource surveys [11], [12], among others.

Many past studies have been dedicated to single-scene classification, which involves the selection of samples within a specific scene for model training and subsequent testing within the same scene. A variety of methodologies have been employed, ranging from traditional machine learning techniques such as k-nearest neighbors [13], support vector machines [14], and sparse representations [15], to more recent deep learning-based methods including convolutional neural network-based [16], graph neural network-based [17] and transformer networks [18]. While these methods have demonstrated commendable performance, a notable limitation is the necessity for ample labeled samples to retrain the model for new scenes. The process of accurately labeling numerous classes within a new scene is both time-intensive and expensive [19]. Furthermore, the resulting model is typically scene-specific, limiting its applicability to the unseen scene. As a result, model generalization to other scenes has drawn wide attention recently.

However, a direct application of a model trained in the source domain (SD) to the target domain (TD) often results in a notable decrease in performance. This is primarily attributed to the disparities in sensor types, diurnal variations, seasonal changes, and meteorological conditions [20] that capture the two scenes, resulting in spectral shifts [21] and spatial texture variations of the same kind of land cover between scenes. To improve the generalization ability of the model to the TD, domain adaptation (DA) techniques have been employed for cross-scene HSI classification [22], [23], [24]. These techniques are often implemented as a form of transductive [25] transfer learning, facilitating the acquisition of shared, cross-domain invariant knowledge between the SD and TD. This, in turn, improves the performance of the model in the TD.

More specifically, the main methods for cross-scene HSI classification based on DA encompass probability distribution alignment [26], [27], [28], feature selection [29], [30], [31], subspace learning [32], [33], [34],

Manuscript received 11 March 2024; revised 21 April 2024; accepted 1 May 2024. Date of publication 15 May 2024; date of current version 29 May 2024. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant 62171451 and Grant 62306323. *(Xi Chen and Lin Gao contributed equally to this work.) (Corresponding author: Maojun Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3401231

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and deep DA [35], [36], [37]. Probability distribution alignment mitigates domain shift by aligning the data distribution between the SD and TD, with maximum mean discrepancy (MMD) [38] frequently employed to quantify the distributional divergence. Feature selection facilitates the extraction of cross-domain invariant knowledge by identifying shared features between the SD and TD. Subspace learning assumes that the SD and TD have similar distributions in the transformed subspace, and DA is realized by minimizing the distributional disparity between the SD and TD within this subspace.

Deep DA employs deep neural networks to automatically extract representations of the SD and TD, achieving cross-domain invariant knowledge learning by aligning data distributions of the SD and TD [39]. For instance, multirepresentation adaptive network (MRAN) [35] align distributions of multiple representations extracted by a hybrid structure to capture information from different aspects between the SD and TD; deep subdomain adaptive network (DSAN) [36] learns to transfer the network by matching the distribution of relevant subdomains activated at specific domain layers between the SD and TD via local MMD; confidence learning-based DA (CLDA) [40] utilizes DA to reduce the interdomain variance and generates pseudo-labels for the target instances, and then selects the high-confidence target samples from the generated pseudo-labels as the training data via confidence learning in order to improve the generalization ability of the neural network in the TD; supervised contrastive learning-based unsupervised DA (SCLUDA) [41] reduces the differences between SD and TD by using a domain similarity loss function based on a single classification task; and a topological structure and semantic information transfer network (TSTnet) [37] leverages graph-optimized transfer to align topological relationships between different land cover classes and assists the distribution alignment between SD and TD through MMD. The class-aligned and class-balanced generative DA (CCGDA) [39] method addresses the challenges of cross-domain distribution differences, class imbalance, and complex spatial–spectral distributions in HSI classification by adopting a hierarchical capsule network classifier, an adaptive sampler, and a novel category-aligned domain adversarial loss function.

Although cross-scene HSI classification methods based on DA have achieved satisfactory classification performance, which not only require data from the TD to participate in the model training but also the obtained model knowledge can only be used for the specific target scenes participating in the model training process. When facing other scenes that do not participate in the model training, the model must be retrained for a specific scene. To liberate the model from the dependence on the TD data, it is desired that the model learns using only the SD data to exhibit better generalization performance in unknown domains, and domain generalization (DG) [42] is a promising direction to address this problem. However, traditional DG (multisource DG) requires multiple labeled SDs during training [43]. It is difficult for cross-scene HSI classification because it is hard to obtain labeled hyperspectral data from multiple scenes with the same class. In contrast, single-source DG aims to train a deep model from a single SD that can be generalized to the unseen domain, which aligns more with cross-scene HSI classification application scenarios in practice. The most prevalent methods to tackle single-source DG are domain extensions [44], [45], [46], [47], [48], [49], which extend a single SD by generating diverse pseudo-domains. For example, the meta-learning-based adversarial domain augmentation (M-ADA) [44] method utilizes adversarial training to create "fictitious" but "challenging" domains. The progressive domain extension network (PDEN) [45] generates multiple domains progressively through a domain extension subnetwork to simulate various photometric and geometric transformations in unseen domains. It introduces contrastive learning to learn domain-invariant representations. Learning to diversity (L2D) [46] for the single-source DG method utilizes a style-complementary module to enhance the model's ability to generalize to the unseen domain by synthesizing images with different distributions complementary to the SD image. An adversarial Bayesian augmentation (ABA) [47] for single-source DG takes full advantage of adversarial learning and Bayesian neural networks to guide the generation of different augmented data to help classifiers generalize to the unseen domain. A simple expansion for single source DG (SimDE) [48] method generates nonconfidence samples by combining entropy maximization and cross-entropy (CE) minimization to learn the desired domain expansion. The single SD extension network (SDEnet) [49] enables a semantic coder to generate extension domains using supervised contrastive adversarial learning strategies. It enables a discriminator to learn class-domain-invariant representations from extension, source, and intermediate domains.

Traditional domain extension methods designed for general images focus only on the changes in spatial information of the images. However, for cross-scene HSI classification, domain differences are brought by spatial texture information changes and spectral drift, causing poor performance in cross-scene HSI classification tasks. In addition, the diversity of the extended domain directly affects the model's generalization ability on unseen domains. Due to the unique 3-D spatial–spectral structure of HSIs, ensuring the diversity of the extended domain while maintaining the essential class semantic information represented by the spatial–spectral properties of the extended domain, not deviating from the SD, will have a direct impact on the model's ability to capture cross-domain invariant class-domain semantic information.

To address the aforementioned challenges, we propose a spectral–spatial adversarial multidomain synthetic network (S^2AMSnet) for cross-scene HSI classification. Specifically, S^2AMSnet consists of two structurally identical but parameter-unshared spectral–spatial domain-generation networks (S^2DGNs), a semantic consistency discriminator, a multiscale mutual information regularization network (MMIRN), and a multidomain mixing module. Two S^2DGNs with identical but not shared parameters generate spatially and spectrally diverse generative domain (GD) and drive the semantic consistency discriminator to capture the essential class semantic information from the diversity of the GD. To constrain the intrinsic class semantic information of the GD from deviating from the SD, the MMIRN is used to maximize the mutual

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information (MI) of the spectral–spatial fusion features of the generative and original spectral patches at different spatial scales to ensure that the intrinsic class semantic information of the GD is consistent with that of the SD. The multidomain mixing module linearly mixes the SD and GD in random proportions to produce a mixed domain (MD) to increase the GD diversity further and enhance the class-domain semantic consistency between the GD and the SD.

The main contributions of this article are summarized as follows.

1) A novel S^{2}DGN is designed to generate a GD with spatial and spectral variations to accommodate spatial variations and spectral shifts in different scenes in the cross-scene HSI classification task.
2) To generate diversity GD, two S^{2}DGNs with the same structure but different parameters are made to generate diversity GDs by the designed two-step min–max training strategy.
3) To ensure the class-domain semantic consistency between the GD and the SD, an MMIRN is developed to constrain the S^{2}DGN so that it generates GD with class semantic information similar to the SD.
4) On three real cross-scene HSI datasets, S^{2}AMSnet achieved competitive results compared to the state-of-the-art methods.

The remainder of this article is organized as follows. Section II briefly describes the related work covered in this article. In Section III, we describe the principle of the proposed method in detail. Then, we verify the effectiveness and superiority of the proposed method through extensive experiments and analysis in Section IV. Finally, we summarize the work of this article in Section V.

II. Related Work

A. Involution

In convolutional neural networks, the 2-D convolution employs kernels that slide along the spatial dimension and share convolutional kernel weights in the spatial dimension [50]. Compared to the convolution operation of involution [51], involution has opposite properties. The shape of the convolution kernel of the involution depends on the shape of the input feature map, and its convolution kernel slides along the channel dimension. Such an operation mechanism is suitable for extracting spectral features from HSIs. Let $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{H\times W\times C}$ denote the input feature map, where H and W denote its height and width, and C denotes the input channel. For a pixel $\boldsymbol{\mathcal{X}}_{i,j} \in \mathbb{R}^C$ at coordinates (i, j), its corresponding involution kernel $\boldsymbol{\mathcal{H}}_{i,j}$ is obtained through the kernel generating function $\Phi : \mathbb{R}^C \to \mathbb{R}^{K\times K\times G}$, which can be formalized as follows:

$$\boldsymbol{\mathcal{H}}_{i,j} = \Phi\left(\boldsymbol{\mathcal{X}}_{i,j}\right) = \boldsymbol{\mathcal{W}}_1\delta\left(\boldsymbol{\mathcal{W}}_0\boldsymbol{\mathcal{X}}_{i,j}\right) \tag{1}$$

where $\boldsymbol{\mathcal{W}}_0 \in \mathbb{R}^{(C/r)\times C}$ and $\boldsymbol{\mathcal{W}}_1 \in \mathbb{R}^{(K\times K\times G)\times(C/r)}$ denote the two linear transformations, r is the channel dimension reduction ratio, G is the number of groups sharing the same involution kernel in each group, and δ denotes the batch normalization and nonlinear activation function.

The output feature map of the involution is obtained by multiply-add operations on the input feature map with the involution kernel, which can be expressed as

$$\mathbf{Y}_{i,j,k} = \sum_{(u,v)\in\Delta_K} \boldsymbol{\mathcal{H}}_{i,j,u+\lfloor K/2\rfloor,v+\lfloor K/2\rfloor,\lceil kG/C\rceil}\boldsymbol{\mathcal{X}}_{i+u,j+v,k} \tag{2}$$

where Δ_K is the set of offsets in the neighborhood convolution conducted on the center pixel, denoted as

$$\Delta_K = \left[-\lfloor K/2\rfloor,\ldots,\lfloor K/2\rfloor\right]\times\left[-\lfloor K/2\rfloor,\ldots,\lfloor K/2\rfloor\right] \tag{3}$$

where $\times$ denotes the Cartesian product.

B. MI Estimation

Information theory commonly uses MI to measure the degree of correlation between two random variables. For MI between random variables X and Y, $\mathrm{MI}(X;Y)$ can be understood as the reduction in the uncertainty of X given Y [52] and can be formalized as

$$\mathrm{MI}(X;Y) = H(X) - H(X|Y) \tag{4}$$

where $H(\cdot)$ is the Shannon entropy, and $H(X|Y)$ is the conditional entropy of Y given X.

However, MI estimation is extremely difficult for continuous and high-dimensional data. Due to the excellent feature extraction and nonlinear representation capabilities of deep neural networks, MI can be estimated by deep neural networks [52], [53], [54]. Oord et al. [55] proposed to estimate MI by optimizing the InfoNCE-type lower bound of MI [56]. Specifically, for the data $\mathbf{X} = \{\mathbf{x}_1,\ldots,\mathbf{x}_N\}$ and $\mathbf{Y} = \{\mathbf{y}_1,\ldots,\mathbf{y}_N\}$, which contains N samples, the $\mathrm{MI}(\mathbf{X},\mathbf{Y})$ between $\mathbf{x}$ and $\mathbf{Y}$ can be evaluated as follows:

$$\mathrm{MI}(\mathbf{X},\mathbf{Y}) \geq \log(N) - \mathcal{L}_{\mathrm{InfoNCE}} \tag{5}$$

$$\mathcal{L}_{\mathrm{InfoNCE}} = \frac{1}{N}\sum_{i=1}^{N}\log\frac{s(\mathbf{x}_i,\mathbf{y}_i)}{\sum_{j=1}^{N}s\left(\mathbf{x}_i,\mathbf{y}_j\right)} \tag{6}$$

where $s(\mathbf{x}_i,\mathbf{y}_i) = \exp(F(\mathbf{x}_i;\theta)^{\mathrm{T}}F(\mathbf{y}_i;\theta)/r)$ is a similarity function to compute $\mathbf{x}_i$ and $\mathbf{y}_i$, $F(\cdot;\theta)$ is a deep neural network with parameter θ, and r is a temperature parameter used to control softness. As N increases, $\mathrm{MI}(\mathbf{X},\mathbf{Y})$ gets smaller and smaller. When $N \to \infty$, it can be obtained that $\log(N) - \mathcal{L}_{\mathrm{InfoNCE}} \to \mathrm{MI}(\mathbf{X},\mathbf{Y})$. Thus, minimizing the InfoNCE loss $\mathcal{L}_{\mathrm{InfoNCE}}$ can maximize the lower bound of MI, thereby estimating MI.

III. Proposed Method

Suppose the SD containing N samples is $\mathcal{S} = \{\mathcal{X}_i,\mathcal{Y}_i\}_{i=1}^N$ and the TD containing M samples is $\mathcal{T} = \{\mathcal{X}_i,\mathcal{Y}_i\}_{i=1}^M$, where $\mathcal{X}_i$ and $\mathcal{Y}_i$ denote the spectral patches with their corresponding class labels, respectively. For the single-source DG setting, only the SD $\mathcal{S}$ is used for model training so that the learned model can be well generalized to the TD. Notably, the TD $\mathcal{T}$ is not accessible during training. To achieve excellent classification performance when models trained only on SD are directly transferred to the TD, S^{2}AMSnet enables two S^{2}DGNs with the same structure but not shared parameters to

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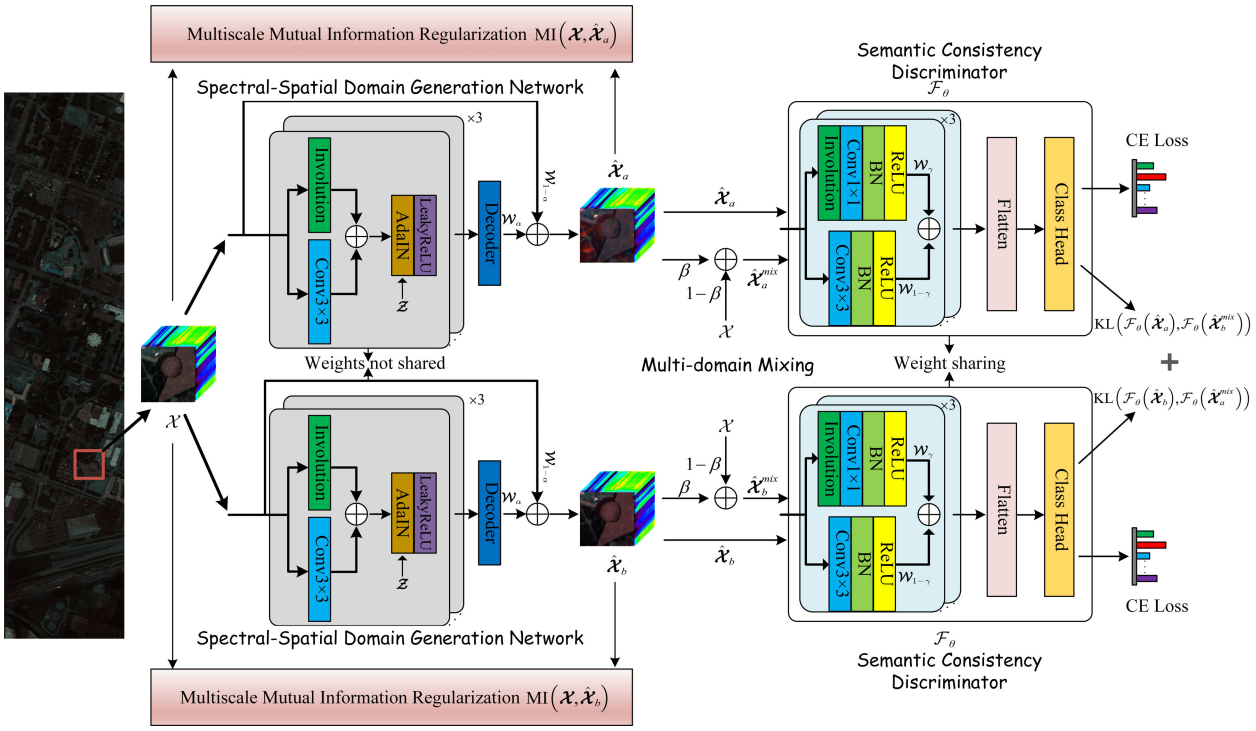


Fig. 1. Schematic of S^{2}AMSnet for cross-scene HSI classification.

generate diverse GD through a two-step min–max adversarial training process, and the semantic consistency discriminator is trained to capture the intrinsic class semantic information from the diverse GD. The GD generated by the S^{2}DGN is passed through the multidomain mixing module to expand the diversity of the GD further. To guarantee the consistency of class semantics between the GD and the SD, multiscale MI regularization constraints between the SD and the GD are used to ensure the consistency of class semantics between the GD and the SD. Diverse and similar to the SD class semantics of the GD can effectively improve the performance of the model on the unseen domain. The schematic of the S^{2}AMSnet is shown in Fig. 1, which includes the S^{2}DGN, semantic consistency discriminator, MMIRN, and multidomain mixing module.

A. Spectral–Spatial Domain-Generation Network

To generate diversified GD to enhance the generalization ability of the model, two S^{2}DGNs with the same structure but not shared parameters are designed in S^{2}AMSnet to generate diversified GD. The model structure of S^{2}DGN is shown as S^{2}DGN in Fig. 1. HSIs have rich spectral and spatial information, and in order to generate diversified GD at the spatial and spectral levels, the S^{2}DGN contains spectral feature branches and spatial feature branches. For an arbitrarily sampled spectral patch $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$, the spectral and spatial features of the spectral patch are first extracted by the spectral and spatial feature branches, respectively. The spectral features $\mathcal{X}^{\text{spe}} \in \mathbb{R}^{H\times W\times B}$ are obtained by involution, which obtains the spectral information of the spectral patch by extracting the spectral features of each pixel in the spectral patch along the band direction. The process can be expressed as

$$\mathcal{X}^{\text{spe}}_{i,j,k} = \sum_{(u,v)\in\Delta_K} \mathcal{H}_{i,j,u+\lfloor K/2\rfloor,v+\lfloor K/2\rfloor,\lceil kG/B\rceil}\mathcal{X}_{i+u,j+v,k} \tag{7}$$

where $\mathcal{H}_{i,j,\cdot,\cdot,g} \in \mathbb{R}^{K\times K}, g = 1, 2, \ldots, G$ is the involution kernel, k is the number of output channels, and G is the number of groups sharing the same involution kernel per group. Δ_K is the set of offsets in the surrounding neighborhood centered on the pixel at (i, j), and B is the number of bands in the spectral patch.

The spatial feature $\mathcal{X}^{\text{spa}} \in \mathbb{R}^{N\times N\times B}$ is obtained by the 2-D convolution on the spatial dimension of the spectral patch, which can be expressed as

$$\mathcal{X}^{\text{spa}}_{i,j,k} = \sum_{c=1}^{C_{\text{in}}} \sum_{(u,v)\in\Delta K} \mathcal{G}_{k,c,u+\lfloor K/2\rfloor,v+\lfloor K/2\rfloor}\mathcal{X}_{i+u,j+v,c} \tag{8}$$

where $\mathcal{G}_{k,c} \in \mathbb{R}^{K\times K}, k = 1, 2, \ldots, C_{\text{out}}, c = 1, 2, \ldots, C_{\text{in}}$ is the convolution kernel for 2-D convolution, C_{out} is the size of the output channel, C_{in} is the size of the input channel, and $C_{\text{out}} = C_{\text{in}} = B$ in S^{2}DGN.

After the spectral feature branch and spatial feature branch, the spectral feature of the spectral patch is fused with the spatial feature to obtain the spectral–spatial fusion feature $\mathcal{X}_f = \mathcal{X}^{\text{spe}} + \mathcal{X}^{\text{spa}}$. The spectral-space fusion features are then subjected to adaptive instance normalization (AdaIN) [57], which can be expressed as

$$\text{AdaIN}(\mathcal{X}_f, \mathcal{Z}) = \text{FC}_1(\mathcal{Z})\frac{\mathcal{X}_f - \mu(\mathcal{X}_f)}{\sigma(\mathcal{X}_f)} + \text{FC}_2(\mathcal{Z}) \tag{9}$$

where $\mathcal{Z} \in \mathbb{R}^{H\times W\times B} \sim \text{Normal}(0, 1)$ is random noise sampled from a normal distribution, FC_1 and FC_2 denote fully

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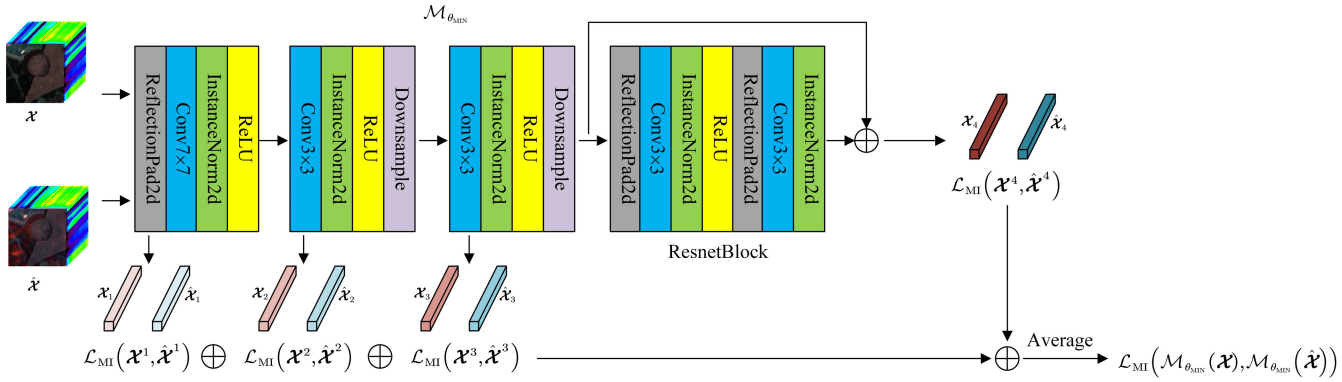


Fig. 2. Schematic of MMIRN structure.

connected layers, and $\mu(\mathcal{X}_{\text{fusion}})$ and $\sigma(\mathcal{X}_{\text{fusion}})$ are the mean and standard deviation of the spectral–spatial fusion features, respectively.

The features output by AdaIN are passed through the leaky rectified linear unit (LeakyReLU) nonlinear activation function and then passed through the decoder to obtain $\tilde{\mathcal{X}}$. Finally, the GD generated by S^{2}DGN is obtained by adaptively fusing $\tilde{\mathcal{X}}$ with the original spectral patch $\mathcal{X}$ through the learnable weights $\mathcal{W}_\alpha$ and $\mathcal{W}_{1-\alpha}$, which can be formalized as

$$\hat{\mathcal{X}} = \mathcal{W}_\alpha \tilde{\mathcal{X}} + \mathcal{W}_{1-\alpha}\mathcal{X} \tag{10}$$

where $\mathcal{W}_\alpha + \mathcal{W}_{1-\alpha} = 1$.

By perturbing the mean and standard deviation of the spectral–spatial fusion features with adversarial training and random noise, S^{2}DGN can generate GD with diversified spatial and spectral information so that the semantic consistency discriminator can capture intrinsic class-domain invariant information from the GD.

B. Multiscale MI Regularized Network

With adversarial training, S^{2}DGN can generate diverse GD. However, if no constraints are imposed on S^{2}DGN, the class semantic information of the GD generated by S^{2}DGN will deviate from the SD, resulting in the semantic consistency discriminator failing to capture class-domain invariant features from the GD. For this reason, the class semantic information of the GD and the SD is constrained by MMIRN between the GD and the SD so that the MI between the GD data and the original spectral patches at different spatial scales is maximized. Inspired by contrastive learning [55], the lower bound of MI [58] between the GD data and the original spectral patches can be estimated by InfoNCE loss [56], which can be formalized as

$$\mathcal{L}_{\text{MI}}(\mathcal{X}, \hat{\mathcal{X}}) = \log \frac{\exp(\hat{\mathcal{X}} \cdot \mathcal{X}/\tau)}{\sum_{\mathcal{X}^- \in N(i)} \exp(\mathcal{X} \cdot \mathcal{X}^-/\tau)} \tag{11}$$

where τ is the temperature parameter, and $N(i)$ is the set of negative samples, $\mathcal{L}_{\text{MI}}(\mathcal{X}, \hat{\mathcal{X}})$ maximizes the semantic consistency between the GD and the SD by maximizing the MI between the original spectral patches and the corresponding GD data.

TABLE I
DETAILED DESCRIPTION OF PARAMETER SELECTION OF MMIRN

MMIRN network structure parameters	Output Dimension
ReflectionPad2D(3)	$\mathcal{X}^1 \in \mathbb{R}^{(H+6)\times(W+6)\times B}$ $\hat{\mathcal{X}}^1 \in \mathbb{R}^{(H+6)\times(W+6)\times B}$
Conv7 × 7(B, 8, 7, 1, 0) InstanceNorm2d(8) ReLU Conv3 × 3(8, 16, 3, 1, 1)	$\mathcal{X}^2 \in \mathbb{R}^{H\times W\times 16}$ $\hat{\mathcal{X}}^2 \in \mathbb{R}^{H\times W\times 16}$
InstanceNorm2d(16) ReLU Downsample(16, 16, 2, 0) Conv3 × 3(16, 32, 3, 1, 1)	$\mathcal{X}^3 \in \mathbb{R}^{H/2\times W/2\times 32}$ $\hat{\mathcal{X}}^3 \in \mathbb{R}^{H/2\times W/2\times 32}$
InstanceNorm2d(32) ReLU Downsample(32, 32, 2, 0) ResnetBlock(32, 32, 3, 1, 1)	$\mathcal{X}^4 \in \mathbb{R}^{H/4\times W/4\times 32}$ $\hat{\mathcal{X}}^4 \in \mathbb{R}^{H/4\times W/4\times 32}$

For a more precise estimation of the MI between the original spectral patches and the GD data, multiscale MI estimation is carried out through the MMIRN. The structure of the MMIRN is shown in Fig. 2, and a detailed description of the MMIRN parameter selection is presented in Table I. The MMIRN first fills the surrounding boundaries of the original spectral patch $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$ and the GD data $\hat{\mathcal{X}} \in \mathbb{R}^{H\times W\times B}$ in the form of symmetric mirrors through the ReflectionPad2D layer to avoid losing the information of the boundary pixels, and obtains $\mathcal{X}^1 \in \mathbb{R}^{H'\times W'\times B}$ and $\hat{\mathcal{X}}^1 \in \mathbb{R}^{H'\times W'\times B}$. Then, the extended original spectral patch $\mathcal{X}^1 \in \mathbb{R}^{H'\times W'\times B}$ and GD data $\hat{\mathcal{X}}^1 \in \mathbb{R}^{H'\times W'\times B}$ are downsampled by the Conv7 × 7-InstanceNorm2d-ReLU module, and the first downsampled features $\mathcal{X}^2 \in \mathbb{R}^{H/2\times W/2\times 16}$ and $\hat{\mathcal{X}}^2 \in \mathbb{R}^{H/2\times W/2\times 16}$ are obtained by the Conv3 × 3 layer with stride 1 and padding. After that, $\mathcal{X}^2 \in \mathbb{R}^{H'\times W'\times 16}$ and $\hat{\mathcal{X}}^2 \in \mathbb{R}^{H'\times W'\times 16}$ are downsampled through the learnable Downsample layer, and the second downsampled features $\mathcal{X}^3 \in \mathbb{R}^{H/2\times W/2\times 32}$ and $\hat{\mathcal{X}}^3 \in \mathbb{R}^{H/2\times W/2\times 32}$ are obtained through the Conv3 × 3 layer with stride 1 and padding. Finally, after downsampling $\mathcal{X}^3 \in \mathbb{R}^{H/2\times W/2\times 32}$ and $\hat{\mathcal{X}}^3 \in$

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$\mathbb{R}^{H/2\times W/2\times 32}$ through the learnable Downsample layer, the third downsampled features $\boldsymbol{\mathcal{X}}^4 \in \mathbb{R}^{H/4\times W/4\times 32}$ and $\hat{\boldsymbol{\mathcal{X}}}^4 \in \mathbb{R}^{H/4\times W/4\times 32}$ are obtained through the ResnetBlock. Notably, the downsample layer contains a ReflectionPad2D layer and a 2-D convolution layer with stride 2 and no padding.

For the GD data $\hat{\boldsymbol{\mathcal{X}}}$ and the original spectral patch $\boldsymbol{\mathcal{X}}$, their multiscale MI after going through the MMIRN can be estimated by minimizing the following equation:

$$\mathcal{L}_{\mathrm{MI}}\left(\mathcal{M}_{\theta_{\mathrm{MIN}}}(\boldsymbol{\mathcal{X}}), \mathcal{M}_{\theta_{\mathrm{MIN}}}(\hat{\boldsymbol{\mathcal{X}}})\right) = \frac{1}{N}\sum_{i=1}^{N}\mathcal{L}_{\mathrm{MI}}\left(\boldsymbol{\mathcal{X}}^i, \hat{\boldsymbol{\mathcal{X}}}^i\right) \quad (12)$$

where $\mathcal{M}_{\theta_{\mathrm{MIN}}}(\cdot)$ denotes the MMIRN, θ_{MIN} is the parameters of the MMIRN, and $N = 4$.

C. Multidomain Mixing Module

Under the constraint of multiscale MI regularization, the S^{2}DGN can generate diverse GD with class-domain semantic consistency. In order to assist the semantic consistency discriminator in learning class-domain semantic consistency from diverse GD, the multidomain mixing module is designed to generate MD to help the semantic consistency discriminator learn class-domain semantic consistency. Specifically, the multidomain mixing module generates more diverse MDs that contain information from the SD but are not identical to the SD by linearly mixing the SD with the GD in random proportions, which further increases the diversity of the GD and enhances its class-domain semantic consistency with the SD, which can be formalized as

$$\hat{\boldsymbol{\mathcal{X}}}^{\mathrm{mix}} = \beta\hat{\boldsymbol{\mathcal{X}}} + (1-\beta)\boldsymbol{\mathcal{X}} \quad (13)$$

where β is a random number randomly sampled from a 0 to 1 uniform distribution, and $\hat{\boldsymbol{\mathcal{X}}}^{\mathrm{mix}}$ denotes the MD.

Since the MD further contains more SD information, the MD obtained through the multidomain mixing module is used as template distributions to make the distributions between the GD and the MD more similar to enable the semantic consistency discriminator to capture the class-domain semantic consistency information from the GD and the MD to learn the class cross domain invariant information. The similarity of the distribution between GD and MD is measured using the KL divergence, which can be expressed as

$$\mathcal{L}_{\mathrm{KL}}\left(\hat{\boldsymbol{\mathcal{X}}}, \hat{\boldsymbol{\mathcal{X}}}^{\mathrm{mix}}\right) = \mathrm{KL}\left(\mathcal{F}_\theta\left(\hat{\boldsymbol{\mathcal{X}}}\right) || \mathcal{F}_\theta\left(\hat{\boldsymbol{\mathcal{X}}}^{\mathrm{mix}}\right)\right) \quad (14)$$

where $\mathcal{F}_\theta(\cdot)$ represents the semantic consistency discriminator and θ is the parameters of the semantic consistency discriminator.

D. Semantic Consistency Discriminator

The semantic consistency discriminator is used to capture the intrinsic class-domain semantic consistency information from the GD and MD. Its structure is shown in the semantic consistency discriminator part in Fig. 1. To capture the spectral features and spatial features of the GD and MD to fit the 3-D structure of the HSI, the semantic consistency discriminator includes the spectral feature branch and the spatial feature branch. The spectral feature branch consists of an involution layer, a Conv1 $\times$ 1 layer with stride 1 and no padding, batch normalization, and ReLU. The involution is used to extract the spectral features. Since the involution cannot change the number of channels of the output features, the Conv1 $\times$ 1 layer with convolution kernel size 1 is used to change the number of channels of the output features in the involution. The spatial feature branch uses the structure of Conv3 $\times$ 3-BatchNorm-ReLU. After spectral and spatial feature extraction, the spectral and spatial features are adaptively fused by learnable weights $\boldsymbol{\mathcal{W}}_\gamma$ and $\boldsymbol{\mathcal{W}}_{1-\gamma}$, and it is noteworthy that $\boldsymbol{\mathcal{W}}_\gamma + \boldsymbol{\mathcal{W}}_{1-\gamma} = 1$. Finally, the adaptively fused spectral and spatial features are fed into the classification head to obtain the probability that the sample belongs to each class.

For the GD and MD, to make the semantic consistency discriminator learn the class-domain semantic consistency from them, the classes of the GD data and the MD data judged by the semantic consistency discriminator should be as consistent as possible with the classes belonging to the corresponding SD data, and the CE loss is used to measure the similarity between the classes of the GD data and the MD data judged by the semantic consistency discriminator and the classes belonging to the corresponding SD data, respectively, which can be formalized as

$$\begin{aligned}\mathcal{L}_{\mathrm{CE}}\left(\hat{\boldsymbol{\mathcal{X}}}, \boldsymbol{\mathcal{Y}}\right) &= -\sum_i \boldsymbol{\mathcal{Y}}_i \log\left(\mathcal{F}_\theta\left(\hat{\boldsymbol{\mathcal{X}}}_i\right)\right)\\ \mathcal{L}_{\mathrm{CE}}\left(\hat{\boldsymbol{\mathcal{X}}}^{\mathrm{mix}}, \boldsymbol{\mathcal{Y}}\right) &= -\sum_i \boldsymbol{\mathcal{Y}}_i \log\left(\mathcal{F}_\theta\left(\hat{\boldsymbol{\mathcal{X}}}_i^{\mathrm{mix}}\right)\right)\end{aligned} \quad (15)$$

where $\boldsymbol{\mathcal{Y}}$ is the one-hot coding of the label, and the labels of the GD data and MD data are identical to the labels of the corresponding SD data.

E. Model Optimization

S^{2}AMSnet is trained using a two-step min–max procedure. For the semantic consistency discriminator, the CE loss of the GD and MD with respect to the corresponding labels of the SD is minimized to ensure that the GD and MD have similar class semantic information as the SD, which can be formalized as

$$\begin{aligned}\mathcal{L}_{\mathrm{CE}} =& \mathcal{L}_{\mathrm{CE}}\left(\hat{\boldsymbol{\mathcal{X}}}_a, \boldsymbol{\mathcal{Y}}\right) + \mathcal{L}_{\mathrm{CE}}\left(\hat{\boldsymbol{\mathcal{X}}}_b, \boldsymbol{\mathcal{Y}}\right)\\ &+ \mathcal{L}_{\mathrm{CE}}\left(\hat{\boldsymbol{\mathcal{X}}}_a^{\mathrm{mix}}, \boldsymbol{\mathcal{Y}}\right) + \mathcal{L}_{\mathrm{CE}}\left(\hat{\boldsymbol{\mathcal{X}}}_b^{\mathrm{mix}}, \boldsymbol{\mathcal{Y}}\right)\end{aligned} \quad (16)$$

where $\hat{\boldsymbol{\mathcal{X}}}_a$ and $\hat{\boldsymbol{\mathcal{X}}}_b$ are the generated GD of two S^{2}DGN with the same structure but different parameters, and $\hat{\boldsymbol{\mathcal{X}}}_a^{\mathrm{mix}}$ and $\hat{\boldsymbol{\mathcal{X}}}_b^{\mathrm{mix}}$ are the MD obtained by mixing $\hat{\boldsymbol{\mathcal{X}}}_a$ and $\hat{\boldsymbol{\mathcal{X}}}_b$ with the SD in a randomized ratio, respectively.

To enable the semantic consistency discriminator to learn class-domain semantic consistency information from the GD and MD, the semantic consistency discriminator is enabled to capture class-domain semantic consistency information from diverse domains by minimizing the similarity between the corresponding distributions of GD and MD obtained by the semantic consistency discriminator, which can be represented as

$$\mathcal{L}_{\mathrm{KL}} = \mathcal{L}_{\mathrm{KL}}\left(\hat{\boldsymbol{\mathcal{X}}}_a, \hat{\boldsymbol{\mathcal{X}}}_b^{\mathrm{mix}}\right) + \mathcal{L}_{\mathrm{KL}}\left(\hat{\boldsymbol{\mathcal{X}}}_b, \hat{\boldsymbol{\mathcal{X}}}_a^{\mathrm{mix}}\right). \quad (17)$$

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The overall optimization goal for the semantic consistency discriminator is

$$\min_{\theta} \mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{KL}}. \tag{18}$$

For the S^{2}DGN and the MMIRN, the S^{2}DGN is enabled to generate diverse GD by maximizing the similarity between the corresponding distributions of the GD and the MD obtained by the semantic consistency discriminator. To guarantee that the class semantic information of the GD does not deviate from that of the SD as much as possible, the multiscale MI regularization constraints are imposed on the S^{2}DGN. The multiscale MI regularization constrains the S^{2}DGN to generate GD with class semantic information similar to the SD by making the multiscale MI between the GD and the SD as large as possible, which can be expressed as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{MI}} = {} & \mathcal{L}_{\mathrm{MI}}\left(\mathcal{M}_{\theta_{\mathrm{MIN}}}(\boldsymbol{\mathcal{X}}), \mathcal{M}_{\theta_{\mathrm{MIN}}}\left(\hat{\boldsymbol{\mathcal{X}}}_a\right)\right) \\ & + \mathcal{L}_{\mathrm{MI}}\left(\mathcal{M}_{\theta_{\mathrm{MIN}}}(\boldsymbol{\mathcal{X}}), \mathcal{M}_{\theta_{\mathrm{MIN}}}\left(\hat{\boldsymbol{\mathcal{X}}}_b\right)\right).\end{aligned} \tag{19}$$

The overall optimization objective of the optimal S^{2}DGN and the MMIRN is

$$\max_{\theta_a, \theta_b, \theta_{\mathrm{MIN}}} \mathcal{L}_{\mathrm{KL}} + \mathcal{L}_{\mathrm{MI}} \tag{20}$$

where θ_a and θ_b are the parameters in two S^{2}DGNs with the same structure but different parameters, respectively.

IV. Experiment Results and Analysis

A. Datasets Description

Experiments were performed on three cross-scene HSI classification datasets: Pavia University and Pavia Center datasets, Houston13 and Houston18 datasets, and Hangzhou and Shanghai datasets, in order to validate the effectiveness of S^{2}AMSnet for the cross-scene HSI classification task aimed at single-source DG. Below are the detailed descriptions of the datasets.

1) Pavia University and Pavia Center Datasets: The Pavia University and Pavia Center datasets were acquired by the Reflectance Optical Spectral Imaging System (ROSIS) Sensor in northern Italy. The Pavia University dataset has a spatial size of 610 × 340 with 103 bands. The Pavia Center dataset has a spatial size of 1096 × 715 with 102 bands. They both have a spatial resolution of 1.3 m, a spectral range of 430–860 nm and cover scenes in urban areas. In the experiment, the Pavia University dataset is used as the SD, the Pavia Center dataset is used as the TD, and the last spectral band of the Pavia University dataset is removed so that Pavia University and Pavia Center have the same number of spectral bands. The pseudo-color images of Pavia University and Pavia Center datasets with ground truth are shown in Fig. 3, which contains the seven consistent classes, as shown in Table II.

2) Houston13 and Houston18 Datasets: The Houston13 [59] and Houston18 [60] datasets were acquired by different hyperspectral sensors on the University of Houston campus and its neighboring areas in 2012 and 2017, respectively. The Houston13 dataset has 144 spectral bands with a range of 380–1050 nm, a spatial size of 349 × 1905, and a spatial resolution of 2.5 m. The Houston18 dataset has 48 spectral bands with a range of 380–1050 nm, captured by the ITRES CASI 1500 sensor, with a spatial size of 1202 × 4172 and a spatial resolution of 1 m. The experiment retained the overlapping area containing similar land cover in Houston13 and Houston18. The spatial resolution of Houston18 was reduced to the spatial resolution of Houston13 through downsampling [61] to obtain an overlapping area with a spatial size of 209 × 955, and Houston13 was extracted 48 spectral bands corresponding to Houston18 were used for experiments. The Houston13 dataset is the SD, and the Houston18 dataset is the TD. The pseudo-color images of Houston13 and Houston18 datasets with ground truth are shown in Fig. 4, which contains the seven consistent classes, as shown in Table III.

3) Hangzhou and Shanghai Datasets: Hangzhou and Shanghai datasets were captured by the EO-1 Hyperion hyperspectral sensor, and 198 bands were obtained after removing undesirable bands. Hangzhou dataset was collected in Hangzhou, a city in the east of China, with a spatial size of 590 × 230, and the land covers both urban and rural areas, which mainly include the West Lake and the Qiantang River, land and buildings as well as plants. The Shanghai dataset is collected in Shanghai, a city in the eastern part of China, with a spatial size of 1600 × 230, covering part of the urban

TABLE II
Classes and Sample Numbers in the Pavia University and Pavia Center Datasets

Class		Number of Samples	
No.	Name	Pavia University (Source)	Pavia Center (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

TABLE III
Classes and Sample Numbers in the Houston13 and Houston18 Datasets

Class		Number of Samples	
No.	Name	Houston13 (Source)	Houston18 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

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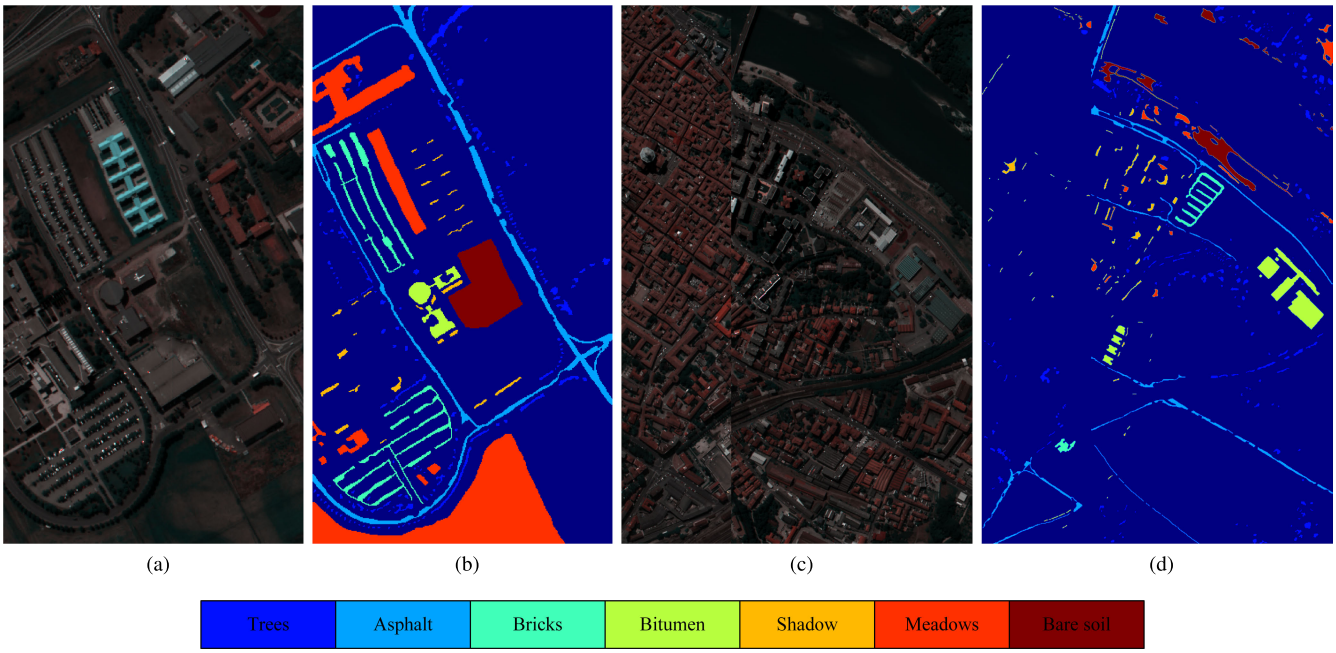


Fig. 3. Pseudo-color images and ground truth for the Pavia University and Pavia Center datasets. (a) Pseudo-color image of Pavia University. (b) Ground truth of Pavia University. (c) Pseudo-color image of Pavia Center. (d) Ground truth of Pavia Center.

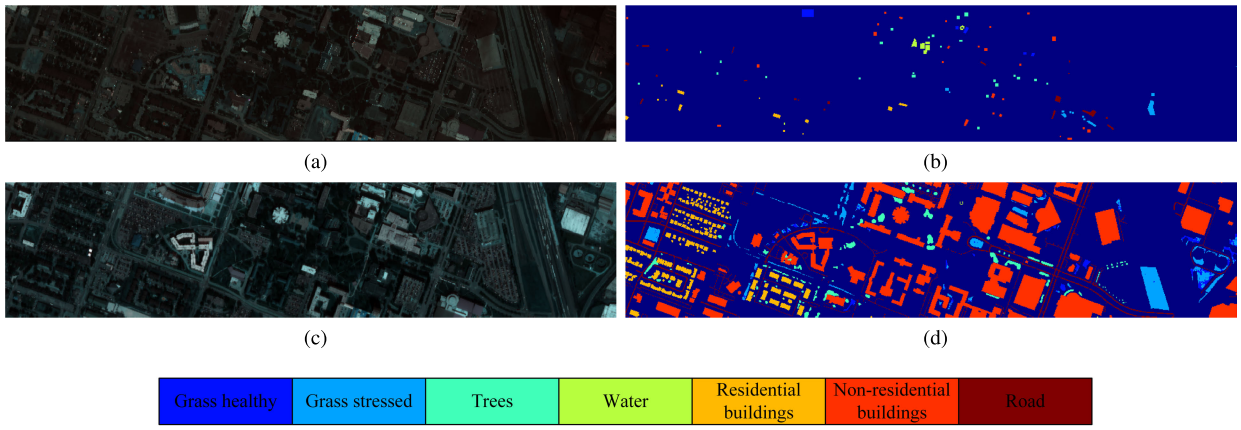


Fig. 4. Pseudo-color images and ground truth for the Houston13 and Houston18 datasets. (a) Pseudo-color image of Houston13. (b) Ground truth of Houston13. (c) Pseudo-color image of Houston18. (d) Ground truth of Houston18.

and rural areas of Shanghai, including the Changjiang River and the Huangpu River, land and buildings, and plants. In the experiment, the Hangzhou dataset is regarded as the SD, and the Shanghai dataset is regarded as the TD. The pseudo-color images of Hangzhou and Shanghai datasets with ground truth are shown in Fig. 5, which contains the three consistent classes, as shown in Table IV.

TABLE IV
CLASSES AND SAMPLE NUMBERS IN THE HANGZHOU AND SHANGHAI DATASETS

Class		Number of Samples	
No.	Name	Hangzhou (Source)	Shanghai (Target)
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40207	83188
Total		135700	368000

B. Experimental Setup

1) Comparison Methods: In the experiments, a total of seven state-of-the-art transfer learning methods are used for comparison with S^{2}AMSnet, including three DA-based methods, MRAN [35], DSAN [36], and TSTnet [37]. Five single-source GD-based methods, learning diversity (L2D) [46] for single-source GD, progressive domain expansion network (PDEN) [45], single-SD expansion network (SDEnet) [49], language-aware DG network (LDGnet) [62], and locally linear unbiased randomization network (LLURnet) [63].

2) Evaluation Criterion: For evaluating the cross-scene HSI classification performance of S^{2}AMSnet against seven state-of-the-art comparative methods, in addition to the qualitative

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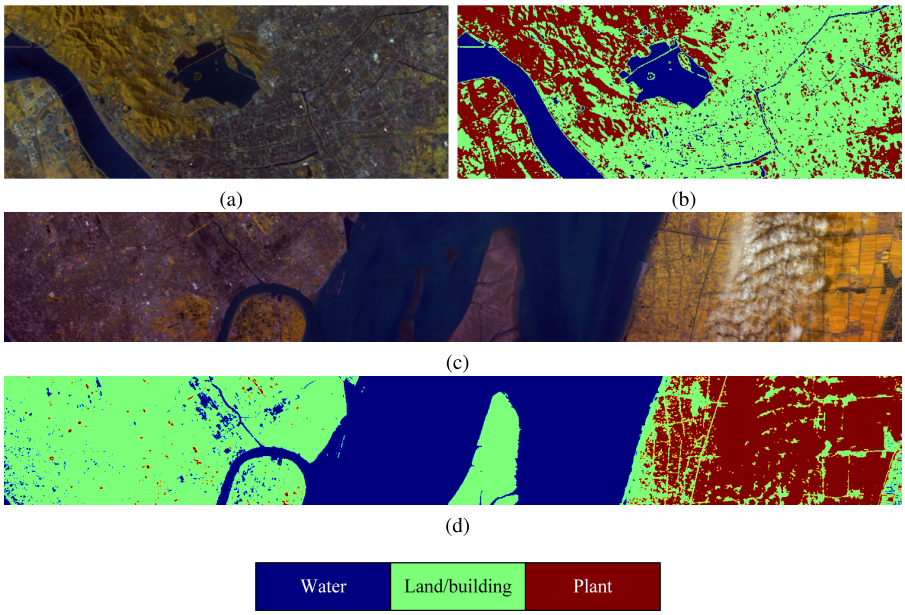


Fig. 5. Pseudo-color images and ground truth for the Hangzhou and Shanghai datasets. (a) Pseudo-color image of Hangzhou. (b) Ground truth of Hangzhou. (c) Pseudo-color image of Shanghai. (d) Ground truth of Shanghai.

analysis based on classification maps, class accuracy (CA), overall accuracy (OA), and kappa coefficient (KC) are used as quantitative evaluation metrics to quantitatively investigate the performance of the proposed method with the comparative methods.

3) Experimental Details: For the sample setting during model training, the DA-based MRAN, DSAN, and TSTnet methods select 80% of the labeled data in the SD with their corresponding labels and all the TD data for model training. The remaining portion of the labeled data in the SD is used as a validation set to verify the model performance. For L2D, PDEN, SDEnet, LDGnet, LLURnet, and S^{2}AMSnet based on single-source DG, only 80% of the labeled data in the SD with their corresponding labels are used for model training, and the remaining 20% of the labeled SD data are used as validation set.

For the parameter settings of the compared methods, DSAN and L2D use Resnet18 as the feature extractor. To adapt to the input size of Resnet18, spectral patches of size 32×32 are used as input. The parameter settings of MRAN, TSTnet, PDEN, PDEN, SDEnet, LDGnet, and LLURnet follow the optimal configurations reported in their original papers.

For the three cross-scene HSI classification tasks in the experiments, the learning rate of S^{2}AMSnet is set to 1e-3. The temperature parameter in the InfoNCE loss function is set to 0.07 when optimizing the InfoNCE loss to estimate the MI between the SD and GD using the MMIRN. The number of downsamples in the MMIRN is set to 3. For the Pavia University and Pavia Center cross-scene HSI classification task, the input spectral patch size for S^{2}AMSnet is 9×9. For the Houston13 and Houston18 cross-scene HSI classification task, the input spectral patch size for S^{2}AMSnet is 13×13. For the Hangzhou and Shanghai cross-scene HSI classification task, the input spectral patch size for S^{2}AMSnet is 7×7.

To reduce the data distribution discrepancy caused by different scales between the SD and TD and enhance model stability, normalization is first applied to the data from both domains during training. In addition, for reliable evaluation, the proposed method and compared methods are performed ten times on each cross-scene HSI classification task in the experiments. The results are averaged to assess the performance of the methods.

4) Experimental Environment: The experimental hardware environment comprises an Intel Core i9-14900k 24-core CPU processor and an NVIDIA Geforce RTX 4090 GPU with 24 GB of RAM. The proposed method and comparison method are implemented using Python 3.8.3 and PyTorch 1.11.

C. Results and Discussion

First, a qualitative evaluation of the classification maps on the target scenes obtained by S^{2}AMSnet and eight state-of-the-art compared methods is conducted visually. Figs. 6–8 present the classification maps of S^{2}AMSnet and the compared methods on the Pavia Center, Houston18, and Shanghai target scenes, respectively, where unlabeled pixels are shown as background and labeled pixels are depicted as the predicted class. The classification maps of S^{2}AMSnet are closer to the ground truth across all three target scenes than the maps of the compared methods. For instance, the second class (asphalt), third class (bricks), and fifth class (shadow), in the Pavia Center, the second class (grass stressed), fifth class (residential buildings), and sixth class (nonresidential buildings) in Houston18, and the second class (land/building) in Shanghai demonstrate considerable improvements over the compared methods in the experiments.

To further quantitatively evaluate S^{2}AMSnet against the competing methods, Tables V–VII report the CA, OA, and KC obtained by S^{2}AMSnet and eight state-of-the-art compared methods on the Pavia Center, Houston18, and Shanghai

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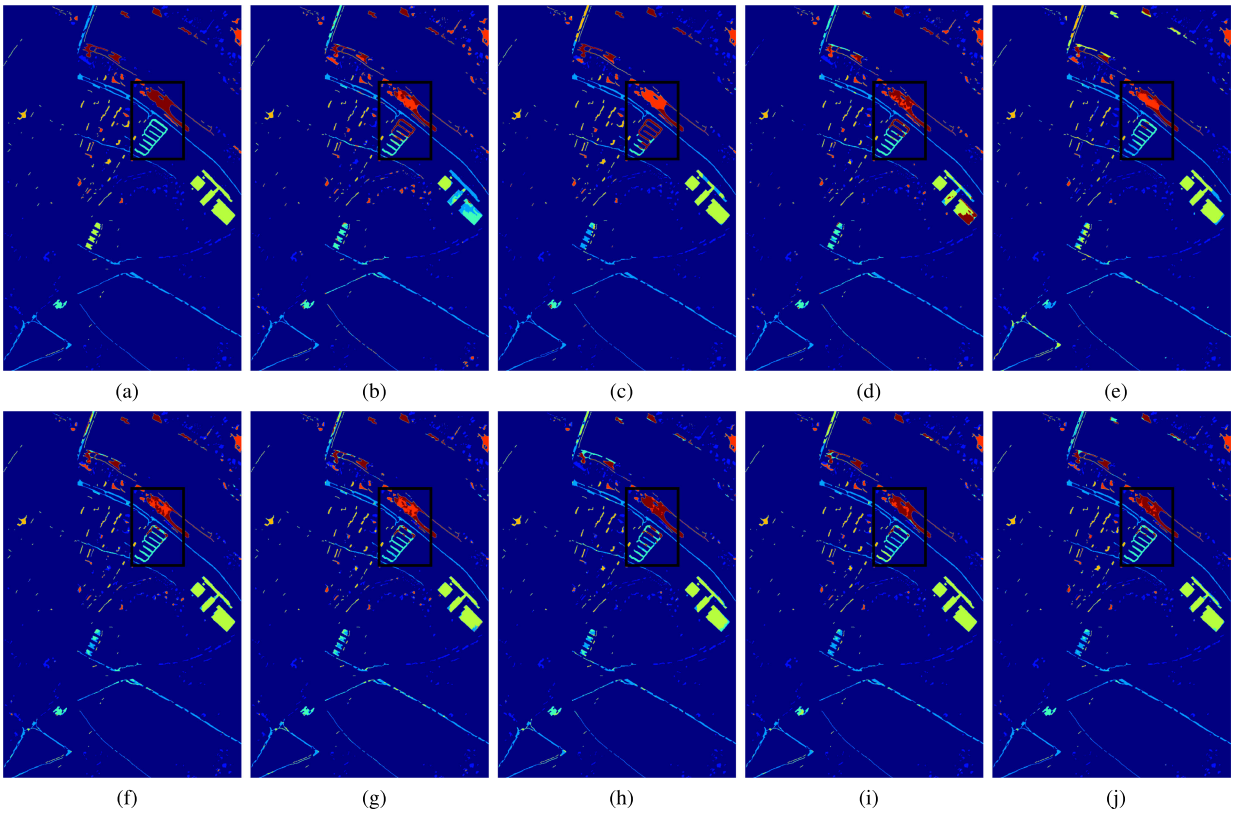


Fig. 6. Classification maps of S^{2}AMSnet and the compared methods on the target scene Pavia Center dataset. (a) Ground truth. (b) MRAN. (c) DSAN. (d) TSTnet. (e) L2D. (f) PDEN. (g) SDEnet. (h) LDGnet. (i) LLURnet. (j) S^{2}AMSnet.

TABLE V
CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF S^{2}AMSNET AND COMPARISON METHODS ON THE TARGET SCENE PAVIA CENTER DATASET

Class	Classification Methods								
	MRAN [35]	DSAN [36]	TSTnet [37]	L2D [46]	PDEN [45]	SDEnet [49]	LDGnet [62]	LLURnet [63]	S^{2}AMSnet
1	69.72	94.96	82.31	93.60	87.08	87.94	97.03	91.70	91.54
2	77.67	89.33	84.14	70.46	81.40	79.04	82.85	87.54	93.06
3	55.75	38.44	68.72	61.53	84.47	66.26	68.60	58.73	90.54
4	18.06	79.21	56.00	85.51	82.16	79.24	76.95	85.93	79.54
5	76.35	92.73	76.63	69.79	82.01	88.02	90.57	92.42	95.53
6	86.41	86.12	86.08	62.33	71.36	75.11	53.30	70.10	74.89
7	71.22	56.23	73.80	53.10	75.96	75.12	79.86	86.42	86.28
OA (%)	62.67±1.62	78.04±0.59	75.62±1.56	72.65±1.84	83.75±0.92	83.72±1.23	81.32±1.09	83.76±1.14	**86.18**±0.61
KC (κ)	55.88±1.96	73.87±0.76	70.83±2.21	66.77±2.53	80.47±1.30	80.39±1.72	77.65±1.32	80.61±1.34	**83.39**±0.92

target scenes, respectively. Based on quantitative evaluation metrics, it can be concluded that in the Pavia Center target scene, for DA-based methods, DSAN outperforms MRAN and TSTnet, and among the five single-source DG-based comparison methods, LLURnet exhibits superior performance. In the Houston18 target scene, for DA-based methods, TSTnet achieved the highest OA, and among the comparison methods based on single-source DG, SDEnet obtained the highest OA. In the Shanghai target scene, TSTnet surpasses the DA-based methods MRAN and DSAN, and LLURnet secures the highest OA among the five single-source DG-based comparison methods. Notably, in the Pavia Center and Houston18 target scenes, the single-source DG-based comparison methods LLURnet and SDEnet outperformed the best DA-based methods, DSAN and TSTnet, by 5.72% and 2.74%, respectively. This indicates that DA methods using domain alignment strategies to reduce interdomain shifts are confined by these strategies, which hinders their capacity to generalize effectively to scenes with complex domain shifts. In contrast, single-source DG methods, which strengthen the model's generalizability by expanding domain diversity in accordance with the characteristics of HSIs, exhibit superior performance

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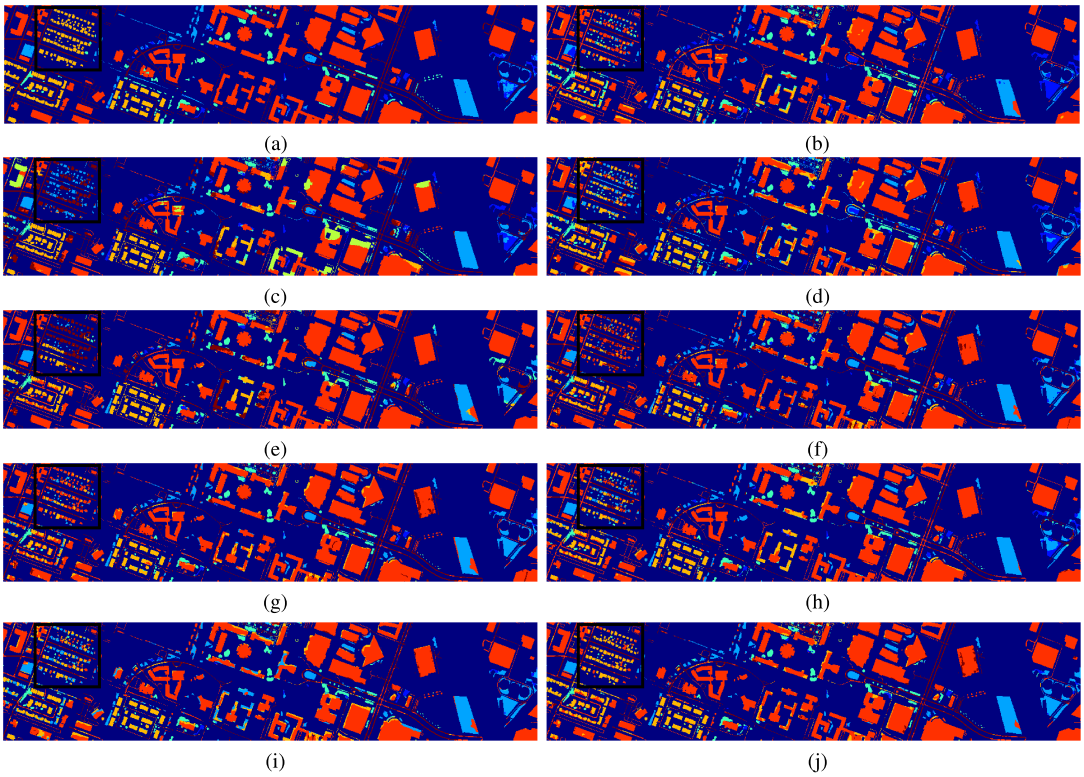


Fig. 7. Classification maps of S^{2}AMSnet and the compared methods on the target scene Houston18 dataset. (a) Ground truth. (b) MRAN. (c) DSAN. (d) TSTnet. (e) L2D. (f) PDEN. (g) SDEnet. (h) LDGnet. (i) LLURnet. (j) S^{2}AMSnet.

TABLE VI

CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF S^{2}AMSNET AND COMPARISON METHODS ON THE TARGET SCENE HOUSTON18 DATASET

Class	Classification Methods								
	MRAN [35]	DSAN [36]	TSTnet [37]	L2D [46]	PDEN [45]	SDEnet [49]	LDGnet [62]	LLURnet [63]	S^{2}AMSnet
1	90.54	64.38	85.22	15.74	40.13	24.54	57.43	32.37	25.42
2	55.18	67.49	81.28	60.88	82.04	78.44	82.82	81.46	90.45
3	72.74	44.65	59.80	47.14	49.28	60.67	63.85	68.26	61.03
4	100	90.91	100	81.82	100	100	81.82	100	100
5	48.68	54.93	69.89	58.29	53.73	50.03	67.18	67.01	80.18
6	91.49	78.87	83.50	91.00	86.25	90.01	91.92	84.84	94.16
7	21.57	54.56	49.55	45.73	62.83	59.92	33.29	48.81	52.77
OA (%)	73.21±1.75	70.59±1.18	76.38±1.94	75.12±1.27	76.74±1.33	79.12±1.10	79.04±0.86	76.82±1.68	**82.70**±0.59
KC (κ)	56.67±1.86	52.74±1.46	62.45±2.05	56.71±2.72	61.21±1.25	63.89±3.77	63.41±2.21	60.58±2.52	**70.16**±0.89

in handling complex domain shifts without leveraging TD data.

The proposed S^{2}AMSnet, designed for single-source DG on cross-scene HSI classification tasks, achieves competitive performance on all target scenes in the experiments. Compared to the best-performing DA-based method in each target scene, S^{2}AMSnet improves the OA by 8.14%, 6.32%, and 2.49% on the Pavia Center, Houston18, and Shanghai target scenes, respectively. Relative to the best-performing single-source DG method in each target scene, S^{2}AMSnet improves the OA by 2.42%, 3.58%, and 3.24% on the three target scenes, respectively. This demonstrates that the S^{2}DGN designed in S^{2}AMSnet can effectively generate diverse GD from spatial and spectral dimensions, the MMIRN can well constrain the class semantic information of GD from the spatial–spectral feature dimensions and the semantic consistency discriminator can successfully capture the class-domain semantic consistency information from the diverse GD with intrinsic class semantic information, thereby realizing a more robust generalization performance.

D. Parameter Analysis

Parameter sensitivity analysis is conducted to evaluate the sensitivity of S^{2}AMSnet on the three cross-scene HSI

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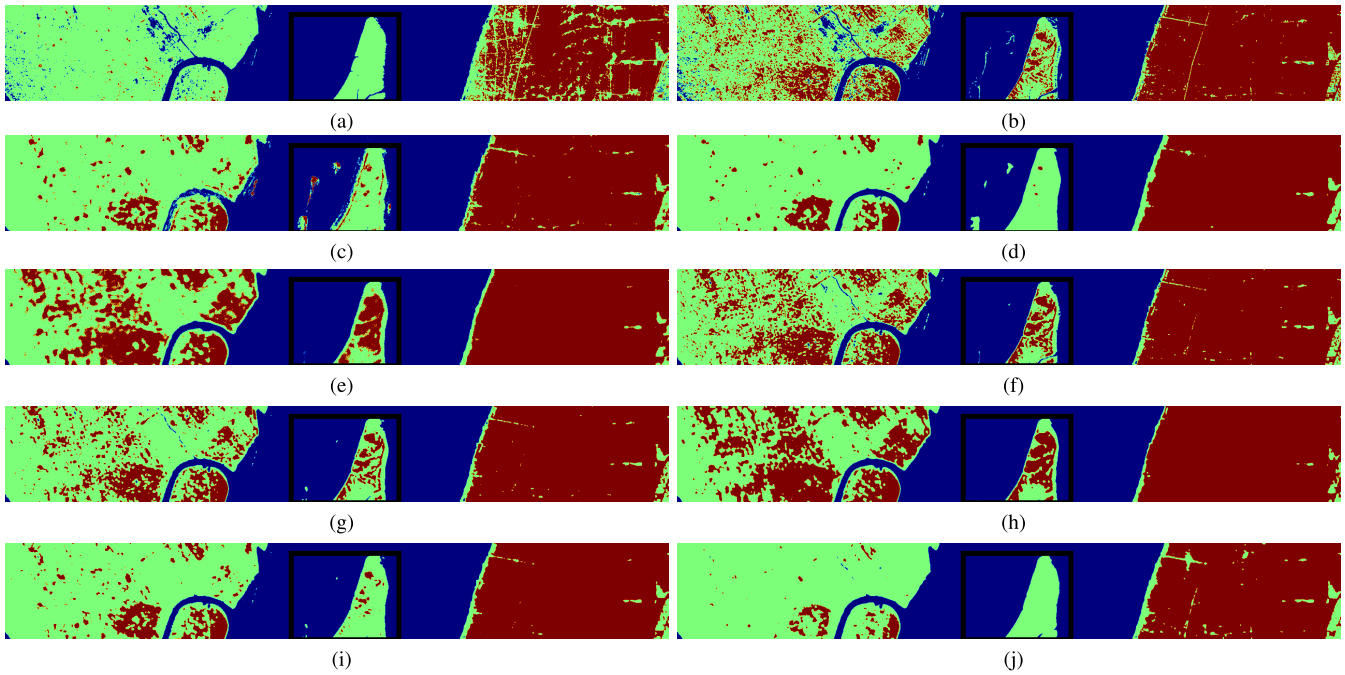


Fig. 8. Classification maps of S^{2}AMSnet and the compared methods on the target scene Shanghai dataset. (a) Ground truth. (b) MRAN. (c) DSAN. (d) TSTnet. (e) L2D. (f) PDEN. (g) SDEnet. (h) LDGnet. (i) LLURnet. (j) S^{2}AMSnet.

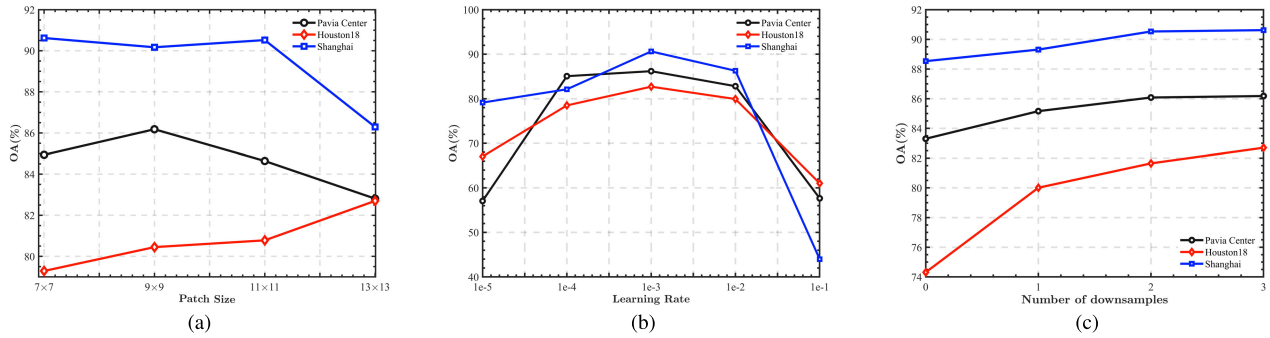


Fig. 9. Parameter sensitivity analysis. (a) Patch size. (b) Learning rate. (c) Number of downsamples.

TABLE VII

CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF S^{2}AMSNET AND COMPARISON METHODS ON THE TARGET SCENE SHANGHAI DATASET

Class	Classification Methods								
	MRAN [35]	DSAN [36]	TSTnet [37]	L2D [46]	PDEN [45]	SDEnet [49]	LDGnet [62]	LLURnet [63]	S^{2}AMSnet
1	96.88	90.16	91.70	93.67	94.87	93.87	94.47	93.44	94.67
2	64.62	77.53	79.60	52.73	58.47	65.73	48.73	76.79	85.99
3	100	99.50	99.42	99.90	99.99	99.95	99.97	99.71	99.12
OA (%)	81.85±1.74	86.22±1.43	88.13±1.28	77.09±1.34	80.61±1.05	81.77±1.99	75.36±1.11	87.38±1.06	**90.62**±1.09
KC (κ)	73.45±2.49	79.48±2.35	82.00±2.13	66.61±2.37	71.39±2.03	73.05±3.94	64.42±2.47	81.07±2.18	**85.56**±2.68

classification tasks. The three major hyperparameters impacting the cross-scene classification performance of S^{2}AMSnet are the patch size, learning rate, and the number of downsamples in the MMIRN.

First, the impact of the spectral patch size on S^{2}AMSnet is investigated. The patch size determines the surrounding spatial neighborhood information utilized for each pixel spectrum. On the three cross-scene HSI classification tasks, four patch sizes of 7 × 7, 9 × 9, 11 × 11, and 13 × 13 are compared. Fig. 9(a) shows the impact of different patch sizes on the cross-scene HSI classification performance of S^{2}AMSnet across the three target scenes. For the Pavia Center target scene, a patch size of 9 × 9 achieves the optimal OA, gradually decreasing as the patch size increases. For the Houston18 target scene, S^{2}AMSnet obtains the highest OA with a patch size of 13 × 13. On the Shanghai target scene, the best results

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TABLE VIII

ABLATION STUDY OF SPATIAL AND SPECTRAL FEATURE BRANCHES IN S^{2}AMSNET ON THREE CROSS-SCENE HSI CLASSIFICATION TASKS (MEAN ± STANDARD DEVIATION)

S^{2}AMSnet		Target Scene OA (%)		
Spatial feature branch	Spectral feature branch	Pavia Center	Houston18	Shanghai
✓	✗	84.28±0.35	76.90±0.99	83.39±0.61
✗	✓	85.11±1.05	79.70±0.98	87.88±1.12
✓	✓	**86.18**±0.61	**82.70**±0.59	**90.62**±1.09

TABLE IX

ABLATION STUDY OF THE MMIRN AND THE MULTIDOMAIN MIXING MODULE IN S^{2}AMSNET ON THREE CROSS-SCENE HSI CLASSIFICATION TASKS (MEAN ± STANDARD DEVIATION)

MMIRN	Multi-domain Mixing	Target Scene OA (%)		
		Pavia Center	Houston18	Shanghai
✗	✗	55.92 ±0.72	63.96±1.85	44.11±0.06
✓	✗	83.50±1.44	80.33±0.34	83.30±2.51
✗	✓	62.10±2.08	67.16±0.74	74.30±1.60
✓	✓	**86.18**±0.61	**82.70**±0.59	**90.62**±1.09

are acquired with a patch size of 11 × 11, and S^{2}AMSnet's performance sharply declines as the patch size becomes larger.

Next, the impact of learning rates of 1e-1, 1e-2, 1e-3, 1e-4, and 1e-5 on the performance of S^{2}AMSnet is evaluated. The learning rate can significantly influence the training process and final results of the network. Fig. 9(b) reports the impact of different learning rates on the cross-scene classification performance of S^{2}AMSnet across the three cross-scene HSI classification tasks. A learning rate of 1e-3 yields the optimal performance for all three tasks for S^{2}AMSnet.

Finally, the impact of different numbers of downsamples in the MMIRN is investigated. The MMIRN captures fused spectral–spatial features at different scales by downsampling the original and generated spectral patches, thereby estimating the MI between them from multiple scales to enforce stricter constraints on class semantics. Fig. 9(c) demonstrates the effect of different numbers of downsamples on the results over the three target scenes. The number of downsamples is 0, which means that the original and generated spectral patches are directly fed into the InfoNCE loss without going through the MMIRN. From Fig. 9(c), it can be seen that the OA of MI estimation on the three target scenes without the MMIRN is significantly lower than that with the MMIRN and the best results can be achieved when the number of downsamples in the MMIRN is 3.

E. Ablation Studies

To generate diverse GD from spatial and spectral dimensions and extract fused spectral–spatial features of HSI, the S^{2}DGN and semantic consistency discriminator in S^{2}AMSnet contain spectral and spatial feature branches. To evaluate the contribution of the spectral and spatial feature branches to S^{2}AMSnet, ablation studies are conducted by removing either the spectral or spatial feature branches from the S^{2}DGN and semantic consistency discriminator to investigate the efficacy of the designed network architecture. Table VIII reports the ablation study results of the spatial and spectral feature branches in S^{2}AMSnet on the three cross-scene HSI classification tasks, where each ablation experiment is repeated ten times, and average results are presented to eliminate the influence of randomness. The OA obtained when only the spatial branch is present is inferior to that of only containing the spectral branch across all three target scenes, indicating the crucial role of spectral features for cross-scene HSI classification. Moreover, the OA attained when both spectral and spatial branches are included outperforms that of only having either branch on the three target scenes, demonstrating the efficacy of the proposed network architecture.

To constrain the S^{2}DGN and prevent deviation in the intrinsic class semantics of the GD from the SD, the MMIRN is designed in S^{2}AMSnet to regularize the class semantics in the GD by estimating the multiscale MI between the GD and SD. In addition, the multidomain mixing module is introduced in S^{2}AMSnet to increase the diversity of the GD further. Ablation studies are conducted on the three cross-scene HSI classification tasks to investigate the efficacy of the MMIRN and multidomain mixing module. Each ablation experiment is repeated ten times to ensure reliable results, and the average results are reported to eliminate the influence of randomness. Table IX presents the ablation study results of the MMIRN and multidomain mixing module in S^{2}AMSnet on the three cross-scene HSI classification tasks. It can be observed that S^{2}AMSnet achieves optimal performance when both the MMIRN and multidomain mixing module are present. Without the MMIRN, the performance substantially decreases, demonstrating its efficacy in constraining the S^{2}DGN to preserve intrinsic class semantics. Removing the multidomain mixing module degrades the performance by 2.68%, 2.37%, and 7.32% on the three target scenes, respectively, which indicates that further increasing the diversity of the GD

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TABLE X
TOTAL NUMBER OF TRAINABLE PARAMETERS AND EXECUTION TIME FOR ONE EPOCH OF TRAINING (IN SECONDS) FOR DIFFERENT METHODS

Methods	MRAN [35]	DSAN [36]	TSTnet [37]	L2D [46]	PDEN [45]	SDEnet [49]	LDGnet [62]	LLURnet [63]	S^2AMSnet
Pavia Center									
Params(M)	0.42	11.57	7.43	11.85	1.57	2.41	35.10	0.58	0.94
Time(s)	86.75	74.39	33.19	139.04	22.23	18.82	58.57	14.14	20.08
Houston18									
Params(M)	0.39	11.41	7.42	11.48	1.48	1.79	34.22	0.55	1.07
Time(s)	19.54	16.91	38.07	19.31	5.67	4.57	18.63	3.14	7.84
Shanghai									
Params(M)	0.43	11.86	7.45	12.50	1.76	3.51	36.60	0.63	1.44
Time(s)	284.65	718.99	258.57	944.43	91.23	70.99	251.13	63.43	68.68

through the multidomain mixing module can increase the model performance, demonstrating the effectiveness of the multidomain mixing module.

In summary, the various modules designed in S^2AMSnet served their purpose.

F. Time Consumption and Model Complexity

To evaluate the computational complexity of S^2AMSnet, Table X shows the total number of trainable parameters and the execution time of training one epoch of S^2AMSnet compared with eight state-of-the-art transfer learning methods. As can be seen from Table X, both the total number of trainable parameters and the execution time of training one epoch of S^2AMSnet are lower than those of domain-adaptive methods. The high time consumption of domain-adaptive methods is mainly due to the fact that the model has to process both the SD data and the TD data, while S^2AMSnet only needs to process the SD data. For DG methods, S^2AMSnet's total number of trainable parameters and the execution time of training one epoch are only superior to L2D and LDGnet and are lower than LLURnet on all three target scenes. S^2AMSnet combines two parameters unshared S^2DGNs and MMIRN to generate diversified and class semantic information undeviated from the SD, although this mode makes S^2AMSnet achieve the best cross-scene classification accuracy, it brings more time consumption.

V. CONCLUSION

The proposed S^2AMSnet is designed for single-source DG on cross-scene HSI classification tasks. It adopts a two-step min–max training strategy where two S^2DGNs with identical architectures but unshared parameters generate spectrally and spatially diverse GD and drive the semantic consistency discriminator to capture class-domain semantic consistency across diverse domains to obtain transferable knowledge for single-source DG. The MMIRN is also introduced to maximize the multiscale MI between the GD and SD, ensuring the intrinsic class semantics are preserved in the GD. The multidomain mixing module further increases the diversity of the GD. S^2AMSnet achieves competitive results compared to state-of-the-art methods on three cross-scene HSI classification datasets, and ablation studies verify the efficacy of each designed module.

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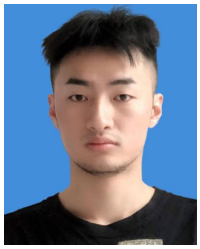
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